

Examining village characteristics for forest management using self- and geographic self-organizing maps: A case from the Baekdudaegan mountain range network in Korea

Tae-Su Kim^{a,1}, Thakur Dhakal^{a,1}, Seong-Hyeon Kim^a, Ju-Hyoung Lee^b, Su-Jin Kim^{c,*}, Gab-Sue Jang^{a,*}

^a Department of Life Sciences, College of Natural Science, Yeungnam University, Gyeongsan 38541, Republic of Korea

^b Department of Forest Resources, College of Life and Applied Sciences, Yeungnam University, Gyeongsan 38541, Republic of Korea

^c Division of Forest Ecology, National Institute of Forest Science, Seoul 02455, Republic of Korea

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ABSTRACT

Understanding the village characteristics linked to forest networks is essential for the scientific management of forest resources. Forests are complex socio-ecological systems. This study classifies the resources and characteristics of forest networks and neighboring villages using unsupervised learning algorithms: self-organizing maps (SOM) and geographic-self-organizing maps (Geo-SOMs). Considering ecological, economic, and socio-cultural indicators, 18 covariates of 379 villages in two forest networks of the Baekdudaegan Mountain Range in South Korea were analyzed. The data visualizing map size was fixed based on changes in quantization and topographic errors of the same grid maps, and the number of clusters was determined by comparing K-means and hierarchical clustering techniques. An optimal map size of 17×12 grids and six clusters was used for further classification of the input data for both SOM and Geo-SOM analyses. The common characteristics of villages were identified using SOM classification, whereas geographically bounded characteristics were identified using Geo-SOM. The approach introduced in this study can be applied to socio-ecological classification and the design of sustainable forest management policies that link the remote sensing and geographic information systems.

1. Introduction

Forests provide environmental, economic, and social welfare benefits through various functions; therefore, national, and regional forest management plans have been established for their management and protection. Forest management in modern society utilizes a forest inventory that provides a statistical explanation of the quality and quantity of forest information collected during various processes (Corona, 2010). Forest inventories provide information for decision-making in forest management and have been applied to solve many environmental problems (Klobucar and Subasic, 2012). Local administrative villages have a significant impact on forest diversity and management (Yang et al., 2021; Ndao et al., 2022). Understanding village characteristics within a complex socio-ecological system supports sustainable forest management (Zang et al., 2021). Village classification and development strategies have both practical and theoretical significance for a green

economy and rural revitalization (Li et al., 2022a; Li et al., 2022b).

Advances in database technology and data mining techniques have made spatial data easier to use and has enabled the discovery of complex relationships, patterns, and their characteristics (Henriques et al., 2009). Clustering, dividing the data into groups of similar objects, following machine learning algorithms, such as hierarchical, partitioning (including K-means, K-medoids, probability-based methods, and density-based methods), grid-based, categorical data-based, constraint, scalable, and high-dimensional data methods (principal component analysis, singular value decomposition, and linear discriminant analysis), are the most efficient ways to solve problems posed by large databases (Berkhin, 2006). Self-organizing map (SOM) is machine learning algorithm proposed by Professor Teuvo Kohonen in the early 1980 s, demonstrating the applicability of unsupervised neural networks to deal with high-dimensional data and optimization problems (Berkhin, 2006).

In studies using regional cluster analysis, the K-means clustering

* Corresponding authors.

E-mail addresses: forestryhydro@korea.kr (S.-J. Kim), sunside@ynu.ac.kr (G.-S. Jang).

¹ These authors contributed equally to this work.

method is frequently used to categorize major variables existing in regions such as mountain villages, rural areas, and cities (Shon and Youn, 1997; Song and Chang, 2010; Kim et al., 2014; Lee, 2014; Jo et al., 2015; Moosavi, 2017; Jeong and Kim, 2019; Ko et al., 2020). K-means clustering method has the advantage of reducing and visualizing high-dimensional data but does not consider topological characteristics (Jeong and Kim, 2019). The SOM preserves the topological relationship of input variables and it can be presented as a lower-dimensional heat map that allows multi-stop analysis and evaluation of clusters that can infer even at the spatial level (Li and Juhola, 2014).

SOMs and their derivatives have been widely used for unsupervised classification in ecological sciences (Chon, 2011; Kim and Kwak, 2022). They have been applied to localization studies, such as forest area classification with SOMs and principal component analysis (Fujino and Yoshida, 2006), spatial clustering with geographic-SOM (Geo-SOM: Bação et al., 2005) landscape classification, and localization studies with geoinformatics and SOMs. Crouzat et al. (2015) clustered 18 ecological parameters, and conducted research from the perspective of ecosystem services using SOMs, while Cao et al. (2013) analyzed the regional characteristics of cities. Fusco and Perez (2015; 2019) studied the statistical development of cities, economies, and social populations by clustering 55 indicators. Clustering using SOMs has also been applied in various fields such as time-series data processing, image compression, land use patterns, vegetation distribution patterns, and hydrological data patterns (Ley et al., 2011; Gao et al., 2014; Liu et al., 2018; Fratarcangeli et al., 2019; Lakshminarayanan, 2020). However, clustering mountain forest networks near villages and environmental features using SOM and Geo-SOM techniques has not yet been demonstrated.

In this study, we applied two forest networks connected to the Baekdudaegan Mountain Range, which is the main mountain range on the Korean Peninsula. These forests have been poorly managed and preserved despite rich resources, which has intensified the damage and disconnection (Choi et al., 2015). Most forest research in The Republic of Korea has mainly been focused on the establishment of mountain range concepts (Korea Forest Service, 1997; Kim et al., 2004) and the conservation management scope of the Baekdudaegan Mountain Range, considering the ecological axis of the Korean Peninsula (Lee et al., 2007; Oh et al., 2007; Shin 2010). Previous studies have focused on the conditions of damaged land (Lee et al., 2007; Jang et al., 2008; Choi et al., 2013; Choi et al., 2015), vegetation (Oh and Park, 2002; Choi and Oh, 2003; Kim and Choo, 2003; Oh et al., 2014; Oh et al., 2016), and ecological networks (Jang, 2007; Jang et al., 2009). Most studies on Jeongmaek, a mountain range forming part of Baekdudaegan, have focused on conserving vegetation (Jang et al., 2008; Lee and Lee, 2011; Park and Kang, 2016; You et al., 2017). To set sustainable management policies, the National Institute of Forest Science (NIFoS) has recently conducted studies on the condition and changes in resources in the mountain ranges, management settings, and resource utilization through a resource-based approach (Kim et al., 2020).

SOMs have been utilized to provide information in a form that can be interpreted intuitively and visually. However, SOMs do not consider geographical features if they have similar characteristics, even if the objects are geographically distant (Bação et al., 2004). Geo-SOM was used to reflect the geographic characteristics of the target site because geographic location can be an important variable when clustering actual spatial data. Therefore, this study aims to support the NIFoS goal by a) visualizing and classifying villages with socio-ecological indicator data using SOM and Geo-SOM and b) analyzing the distribution of indicators in each cluster. The status of land cover and ecological, economic, and socio-cultural indicators were collected to explore the characteristics of administrative villages adjacent to forests. Various regional characteristics, including humanistic and natural environments, were classified and grouped to identify the unique characteristics of each village. The scope of this study was to support policy design for the conservation and management of mountain forests and sustainable ecosystems.

2. Material and methods

2.1. Study area

The Republic of Korea (33–39°N, 124–132°E) is located in East Asia, has an area of 100,210 km², and is located in the mid-latitude temperate zone with four distinct seasons. The forested land ratio is 63.2%; among the Organization for Economic Cooperation and Development (OECD) countries, it is ranked fourth in the forest-to-land area ratio after Finland (73.1%), Japan (68.5%), and Sweden (68.4%; FAO, 2020).

The Baekdudaegan Mountain Range, which is the ecological axis of the Korean Peninsula, is a unique geographic recognition system. It is an approximately 1,400 km-long central mountain range that extends from Mt. Baekdu in North Korea to Mt. Jiri in Korea. This study focused on two forest networks in the 13 mountain ranges forming the Baekdudaegan Mountain Range, Nakdong-Jeongmaek (ND), and Naknam-Jeongmaek (NN), which are located in the southeastern part of the Korean Peninsula (Fig. 1). Various resources are distributed in and around the forest; for our analysis, 379 villages within a 1 km buffer from the ridges of ND and NN were included. Digital topographic maps provided by the Ministry of Land were utilized to determine village boundaries. The data were processed using ArcGIS Pro Ver.2.8.0 (Esri Inc.). At the target site of this study, villages within the area are the main agents of forest management and preservation because of the various resources possessed by the region. A village is a basic social unit, giving its members a sense of solidarity and identity, sharing common values and goals based on a social network (Korea Research Institute for Human Settlements, 2016; Jeonbuk Institute, 2019). Therefore, this study proposes a method for forest management by classifying the common unique characteristics of each village based on socio-ecological, economic, and cultural resources in the mountain range.

2.2. Data

We collected as much forest and village link data as possible to capture the effect of socioeconomic and ecological factors. The data collected from the target site were obtained from reliable sources provided by national and public institutions, including the Korea Forest Service Government, Ministry of Land, Ministry of Environment, Korea Tourism Organization, Cultural Heritage Administration, and municipality websites. After seeking advice from experts and local collaborators, we selected 18 independent covariates: six ecological, five economic, and seven sociocultural classes of 379 villages (Table 1). Forest area (FR), wetland area (WT), water area (WA), forest area above five years age group (F5), protected area (PA), natural environment conservation forest area (NF), agriculture area (AG), water yield forest area (WF), timber production area (TF), recreation forest area (RF), built-up area (BA), living environment forest area (LF; measured within 1 km buffer from the ridge line of the mountain of corresponding villages), distance from living infrastructure (LI), and medical infrastructure (MI) were measured at the center of the village grid, and productive population (human population greater than 21 years old; Pop), cultural heritage (CH), scenic spot (SC), and tourism spot (TS) of each village-connected forest were considered in this study.

2.3. Self-organizing map (SOM)

SOM, a type of ordination map, is one of the unsupervised classification techniques which enables nonlinear data mining with data patterning, clustering, and visualizing properties (Park et al., 2003). SOM is a type of artificial neural network (ANN)-based machine learning technique for visualizing high-dimensional data onto a low-dimensional grid (Kohonen, 1990; Kohonen, 2012), also called Kohonen networks or Kohonen feature maps. SOM networks consist of neurons arranged in a regular lattice structure with rectangular or hexagonal topology (Arribas-Bel and Schmidt, 2013).

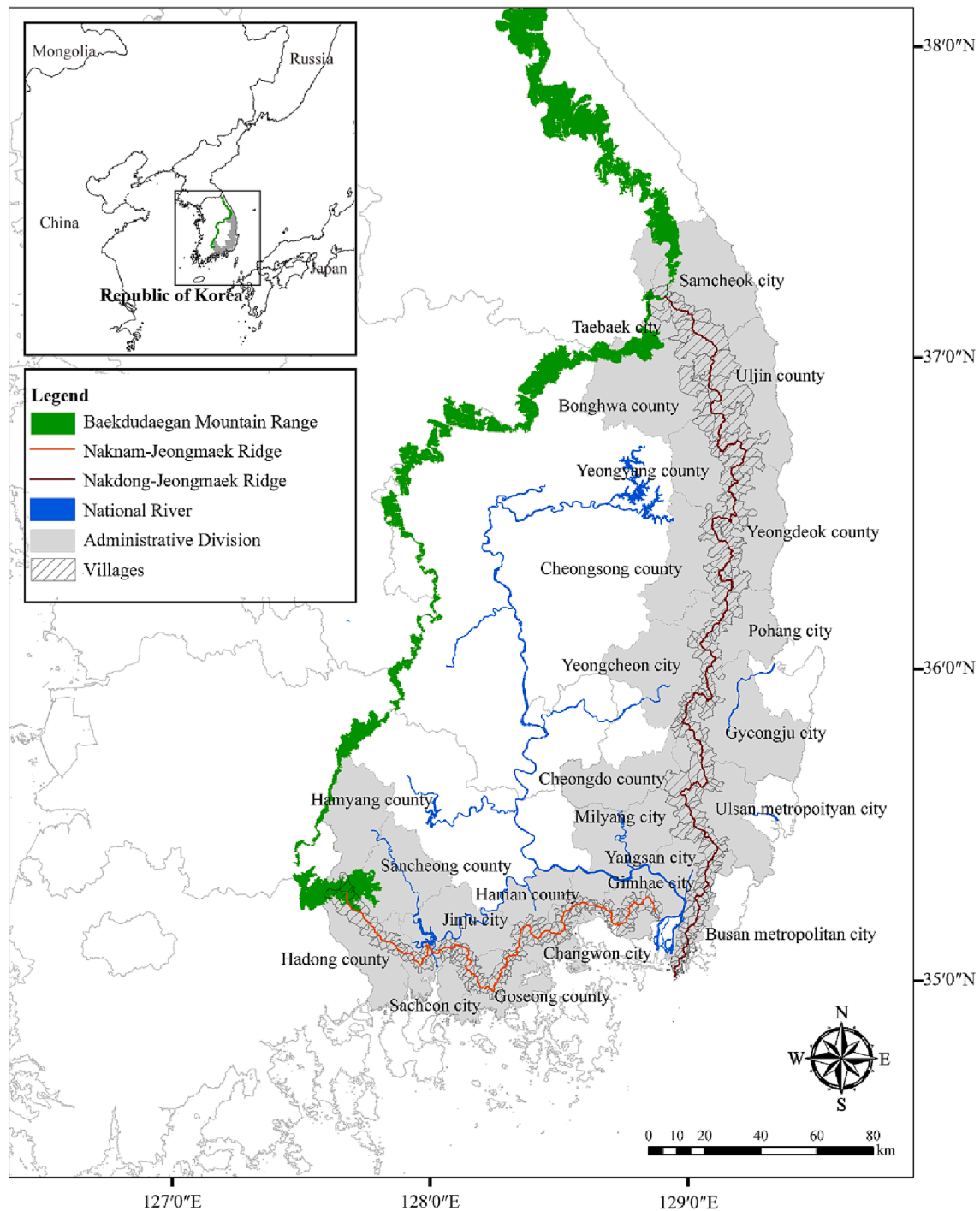


Fig. 1. The map shows the location of the survey region (Korea: rectangular inset in top left) and neighboring countries. All background maps are licensed under CC 4.0.

The SOM was configured with inputs $x_1, x_2, \dots, x_n$, and outputs $o_1, o_2, \dots, o_m$ consisting of m units on a flat grid and a weight vector w_{ij} . Each output contained computational units called neurons or computation nodes. The input data were vector datasets of the corresponding environmental variables. The high-dimensional input datasets are displayed sequentially in a lower-dimensional pattern in the SOM output. The weight vector was randomly set to a small value, and the minimum distance from each weight vector and sample vector was calculated to find the winner neuron, also called the best match unit (BMU), of the SOM. SOMs iteratively run the analysis to bring the BMU closer to the input data. The process continues up to a set number of iterations or ends

when no further executions are performed (Equation (1)).

$$w_{ij}(t+1) = w_{ij}(t) + \alpha(t)h_{ijv}(t)[w_{ij} + x(t)] \quad (1)$$

where $\alpha(t)$ is the learning rate, which decreases with an increase in the repetition step (t), $h_{ijv}(t)$ is the neighborhood function, which defines the distance between adjacent units v and BMU ij . For statistical accuracy, the number of repetitions should be more than 500 times the number of network units, and the number of input variables should not affect the number of repetitions (Kohonen, 2001). The quality of the SOM learning results is generally evaluated using quantization error (QE) and topographic error (TE). QE is calculated using the average

Table 1
Data details.

Indicators	Variables	Unit	Data type	Type of map	Reference
Ecological	FA, WT, WA	km ²	Vector	Land cover map	Ministry of Environment
	F5	km ²	Vector	Forest type map	Korea Forest Service
	PA	km ²	Vector	Ecological nature map	Ministry of Environment
	NF	km ²	Vector	Forest functions classification map	Korea Forest Service Government
Economic	AG	km ²	Vector	Land cover map	Ministry of Environment
	WF, TF, RF	km ²	Vector	Forest functions classification map	Korea Forest Service Government
	POP	Person	Raster	National statistics map	National Geographic Information Institute
Socio-cultural	LF	km ²	Vector	Forest functions classification map	Ministry of Land
	BA	km ²	Vector	Land cover map	Ministry of Environment
	CH	Count	Point	–	Cultural heritage Administration
	TR, SC	Count	Point	–	Korea Tourism Organization
	LI, MI	km	Raster	National statistics map	Ministry of Land

FA: forest area; WT: wetland area; WA: water area; F5: above 5age class forest area; PA: protected area; NF: Natural environment conservation forest area; AG: agriculture area; WF: water yield forest area; TF: timber production forest area; POP: productive population; LF: living environment forest area; BA: built-up area; CH: cultural heritage; TR: tourism spot; SC, scenic spot; LI: distance from living infrastructure; MI: distance from medical infrastructure.

distance between each data input and the BMU to measure the resolution of the map, whereas TE calculates the proportion of all data vectors in which the first and second BMU are not adjacent to measure the topology conservation. Therefore, TE is used as an indicator of mapping accuracy in topology preservation and QE is used to select the best map with the minimum value (Kohonen, 2001; Park et al., 2003).

The SOM adjusts the samples of output units based on the similarity of input variables; however, it is difficult to differentiate between subsets because there are no boundaries between output units. Thus, additional steps were required to classify the map into different groups. The unified distance matrix algorithm (U-matrix) can further cluster SOM-classified BMUs (Siemon and Ullsch, 1990). The U-matrix is inserted between the computed nodes and represents clusters by visualizing the computed distances between the adjacent units. The possible number of clusters within the SOM U-matrix was examined by comparing K-means clustering with the Davies-Bouldin index (DBI; Davies and Bouldin, 1979) and hierarchical clustering through dendrogram approaches. Smaller DBI indicates better clustering and a small DBI value indicates a low variance within the cluster (Hruschka and Natter, 1999; Recknagel and Michener, 2017).

Our data contained three sets of independent indicators (ecological, economic, and socio-cultural); the modified SOM known as a super self-organizing map (supersom) tool in the Kohonen package in the RStudio environment were used. Super SOM can process multiple layers for different classes of input data, and each layer can be viewed as a subset of the training dataset (Wehrens and Buydens, 2007). In other words,

different weighted values can be assigned to each layer by grouping the input variables into layers, and the distance value of each layer can be computed, whereas learning can be biased based on weighted values (Lakshminarayanan, 2020). The purpose of these subsets is to gather a predefined number of input vectors to reduce redundancy. The input vector in SOM provides a map directly, whereas in the supersom, the elements of the input vector contain a predetermined number of elements before providing the map. Supersom can handle missing or non-applicable values by removing records before training the map (Wehrens and Buydens, 2007). In this study, the same weighted value was assigned to all three layers and the data were trained for 5,000 iterations.

2.4. Geographic self-organizing map (Geo-SOM)

Geo-SOM is an extended algorithm of SOM (Bação et al., 2004). The SOM does not consider the characteristics of the geographic location when clustering geo-reference data (Rivas-Tabares et al., 2020), whereas Geo-SOM applies the first law of geography, “Everything is related to everything else, but near things are more related than distance things” (Tobler, 1970) to SOM, preferentially applying neurons that are geographically close to a specific data pattern. Specifically, only units with geographic coordinates similar to the corresponding data are considered to be the best matching units (BMU) when Geo-SOM maps geo-reference data.

The algorithm was first constrained by geographical coordinates, and the other attributes were examined later (Henriques et al., 2009). In the Geo-SOM, the BMU preferentially considers the geographic location of the data and then a final BMU candidate according to the data characteristics of the output space is created. Geographic tolerance is defined on the grid of the SOM and corresponds to shorter distances in regions with higher data densities and longer distances in regions with lower data densities. In the current study, the MATLAB-based Geo-SOM Suite tool was used to analyze the Geo-SOM (Vesanto et al., 2000). The number of clusters was set identical to the SOM.

2.5. Parameters, map size of SOM and Geo-SOM, and significance

Input parameters such as the number of neurons, type of topology, and map size should be set in the initial stage of SOM analysis (Klobucar and Subasic, 2012). We followed the study by Kohonen (2012) to select topology and hexagonal patterns that have more connected neighbor cells than triangular and rectangular patterns. The SOM size used to detect data deviation may vary depending on the number of samples (Kaski, 1997). Map sizes that are too small or too large may influence output performance (i.e., underfitting or overfitting, respectively; Wilppu, 1997). The method of data segmentation into features is understood and tested in supervised learning, but is unknown in unsupervised learning (Dy and Brodley, 2000). Setting the size of unsupervised SOM grids is still unclear. The heuristic approach for selecting the number of neurons is $5 \times \sqrt{\text{number of samples}}$ (Vesanto, 1999; Céréghino and Park, 2009; Recknagel and Michener, 2017; Fortela et al., 2020) and considering QE and TE values (Park et al., 2003). These are the most commonly used methods for the selection of map size. As a map size increases, QE and TE steadily decrease; therefore, the optimal size is fixed based on the local minimum values for QE and TE (Klobucar and Subasic, 2012; Vesanto et al., 2000).

There are no reference values for QE and TE because they depend on the map resolution (Astel et al., 2013). In this study, we attempted to use both the heuristic and error methods to determine the possible number of grids in the SOM. To track the error measures, QE and TE were estimated with different map size combinations $i \times k$ ($i, j = 5, 6, 7 \dots 19, 20$) using the AweSOM package (a tool used to examine the quality of SOM), and minimum changes in errors in the same grid maps were calculated. To determine the difference in the errors in grids, we used the upper

triangular error matrix ([Error matrix]-transpose [Error matrix]), and the minimum error range sizes were obtained from the default set of the surface plot. Minimum error ranged maps were considered and visually compared to the output of 'supersom' of the 'Kohonen' package in RStudio Ver.4.1.1 platform (Rstudio Team, 2021). The performance of the SOM was visually judged based on the scatter of data and the presence of at least four samples in a winner neuron. The optimal map size obtained from the supersom was also considered the optimal map for Geo-SOM analysis in this study. The initial training parameters h , α , and ν for both SOM and Geo-SOM were set to their default values (Henriques et al., 2009; Wehrens and Buydens, 2007).

There is no specific output in unsupervised learning (Dy and Brodley, 2000). In this study, we compared the land features and physical dominance indicators in each cluster by drawing a polygonal plot for both SOM and Geo-SOM, and analyzed the significance of the village classification. The systematic workflow to the current study is summarized in Fig. 2.

3. Results and analysis

A total of 18 covariates were considered to classify 379 villages in the Baekdudaegan Mountain Range for sustainable forest management. Data visualization and patterning were performed using the SOM and Geo-SOM. Map size determination was applied to both SOM and Geo-SOM, and classification analysis was conducted.

3.1. Selection of optimal map size and clusters

The input data consisted of 379 sites (villages) and 18 variables for 6,822 samples. The number of map grids using the heuristic approach was approximately 413, which was too large and inappropriate. We analyzed the QE and TE by setting the width and height of the map with different combinations of sizes ranging from five to 20 and observed that

the errors decreased with increasing map size. The upper triangular deviation of the errors of the same grid map is visualized in a surface plot (Fig. 3). From the surface plot of the TE deviation data, the lowest range was from -0.1 to -0.05 , related to six map sizes of 20×5 , 14×10 , 19×10 , 14×11 , 17×12 , and 19×16 . Similarly, from the QE deviation surface plot, the lowest range was -0.04 to -0.2 , related to five different map sizes of 15×11 , 16×12 , 20×9 , and 20×15 . Ten different maps, including TE (6) and QE (4), were inspected individually. We obtained better grouping performance on a map size of 17×12 (204 grids). This was used throughout the study.

The trained SOM and Geo-SOM designate output samples based on the similarity of the input variables, which are further classified. In our study, DBI for K-means clustering was observed at a minimum of five to seven clusters. Finally, we compared the DBI with a hierarchical dendrogram and set six optimal clusters for further analysis.

3.2. Classification using SOM

The SOM output was visualized in a population heatmap with six identified clusters (Fig. 4a). The distributions of the variables in each group were expressed in code plots (Fig. 4b). The component plans (property maps) for each observed variable from the SOM are shown in Fig. 5. In the SOM obtained from the Kohonen package, the grids were counted from the bottom left to the right, and the final grid ended in the top right cell. The default color of the map was red for minimum values, white for maximum values, and gray for missing values. The thick black lines represent the cluster boundaries (Fig. 3a and 4). The property map (Fig. 5) shows that all variables behaved uniquely, but timber production forests and recreation forests in Cluster II (Fig. 4a) had identical characteristics and higher frequency data.

This study aimed to classify the main characteristics of villages around the forest and focus on forest resources, population density, and the degree of urbanization around the forest. Through SOM analysis, six

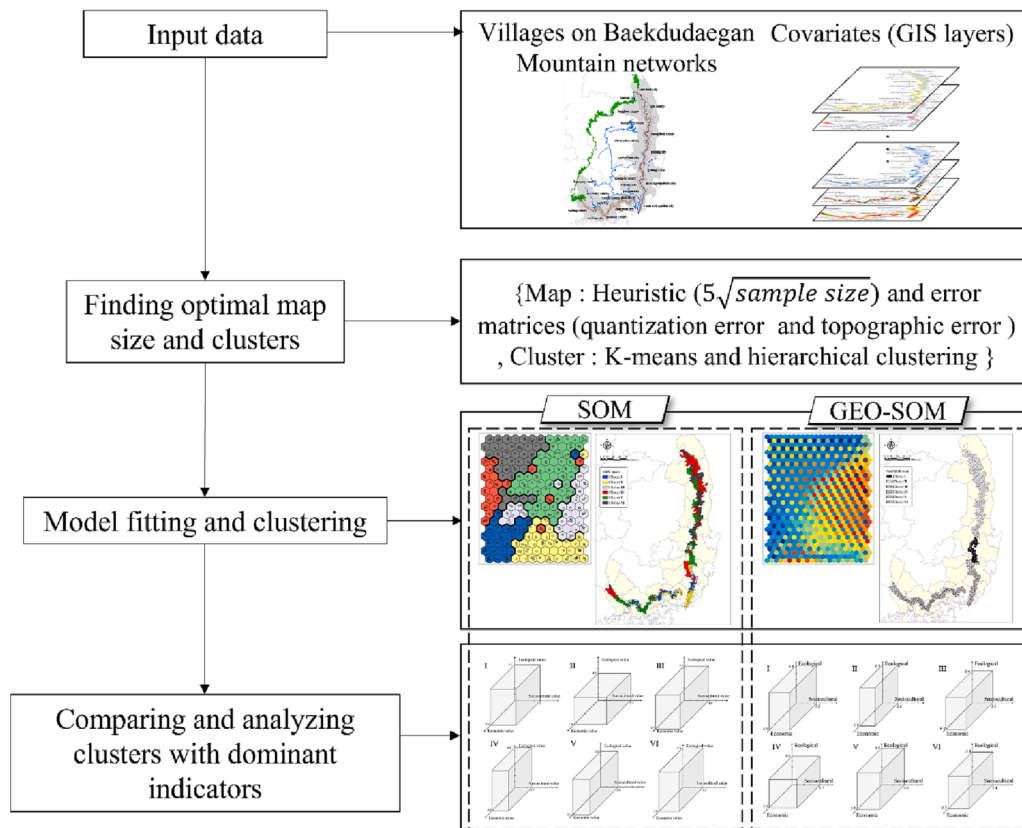


Fig. 2. Study flow.

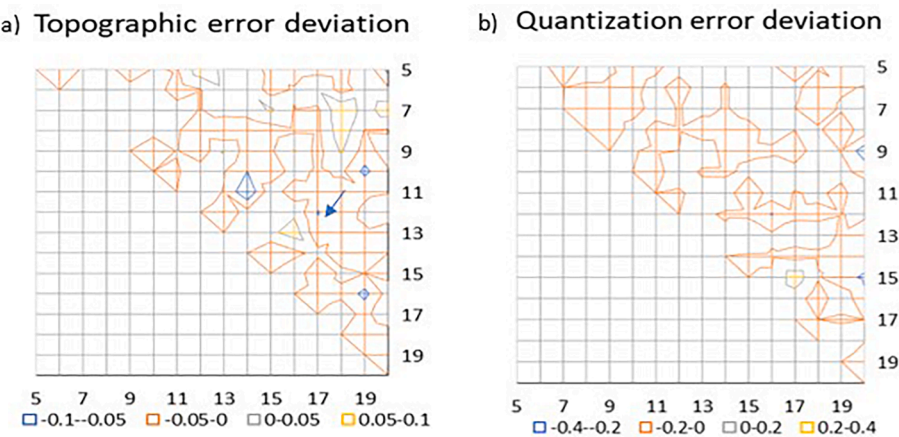


Fig. 3. Error difference surface plot of $(i \times j)$ and $(j \times i)$ maps. a) Topographic and b) quantization error deviation.

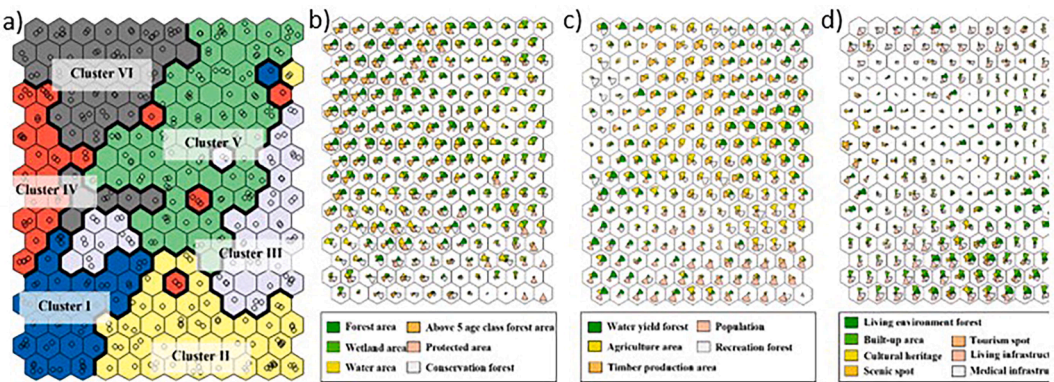


Fig. 4. Data visualization in self-organizing maps (SOMs). a) SOM heat map including clusters and distribution of b) ecological, c) economic, and d) socio-cultural features within the best match units (BMUs).

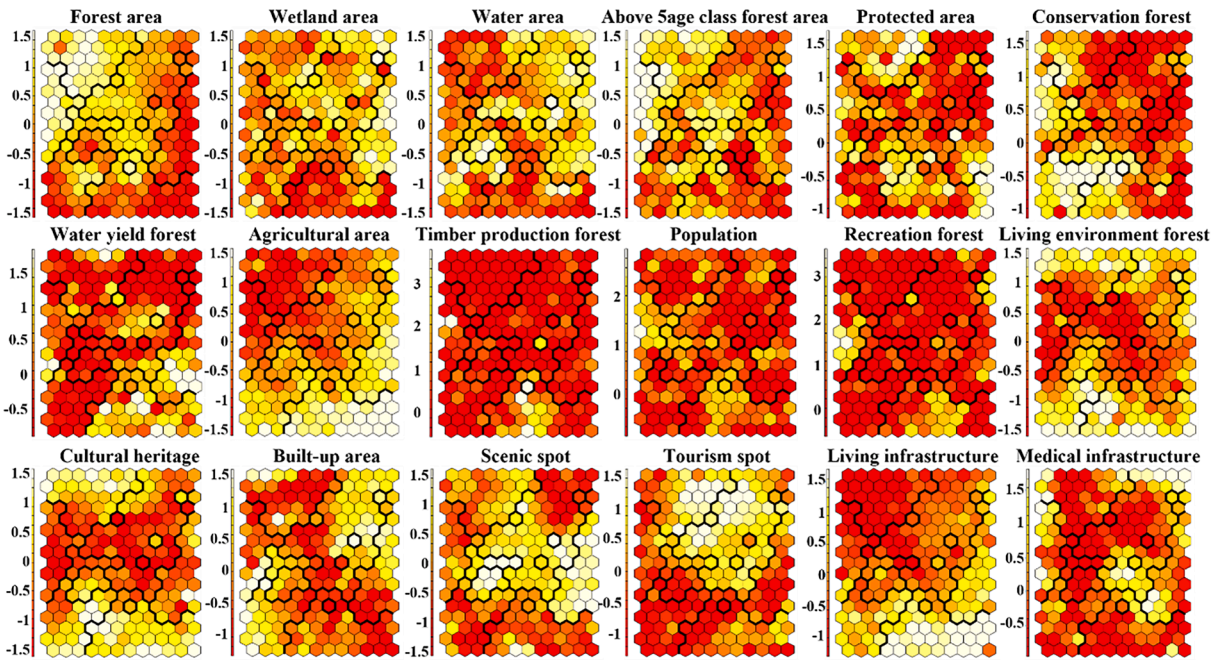


Fig. 5. Visualization of component heat maps.

clusters were classified with the highest number of villages in Cluster V (100), followed by clusters VI (66), II (65), III (57), and I (50), with the lowest number of villages in Cluster IV (41). Considering the aggregation of village properties with the group axis of clusters, a greater ecological value strength in Cluster VI, economic value in Cluster V, and socio-cultural value in Cluster II were identified. Clusters I, III, and IV exhibited complex characteristics (Fig. 6). The average distributions of ecological, economic, and sociocultural indicators in each group were visualized using a polygonal graph that eased the identification of the characteristics of the variables included in the SOM clusters (Fig. 6).

Cluster I villages had the highest NF and WF values and the lowest LF, SC, and TR values. The sociocultural value was low, but LI and MI were high. The overall higher weight of LI and MI values in Cluster I are due to the adjacent large cities, such as Busan, Changwon, and Gimhae, and showed regional characteristics where residential-oriented development or urbanization is in progress. These factors are supported by a highly productive population value.

In Cluster II, socio-cultural values such as POP, BA, CH, MI, and LI seem to have the greatest influence. In contrast, the ecological value and forest-related variables, such as WT, WA, FA, WF, and RF, had the lowest values, indicating that the village of Cluster II represents the characteristics of a typical metropolis. Most villages are distributed in Busan, Gimhae, and Changwon, and the area has been urbanized to address the high population and development pressure. Busan, Korea's second largest marine transportation city on the southeastern coast, is located at the bottom of the ND mountain range, and urbanization has occurred over a period of time. Gimhae and Changwon, located on the far right of the NN mountain range, have also recently undergone rapid urbanization under the influence of neighboring Busan.

Cluster III had the highest AG, WT, and WA values and the lowest FA value. The smallest forest area is composed of paddy fields, and rivers are formed around cultivated land. A high BA value indicates that there are many large-scale industrial complexes in Cluster III. Cluster III shows

the characteristics of agricultural land and industrial complexes on the outskirts of cities and *peri-urbanization*, a characteristic of European cities proposed by the National Institute of Statistics and Economic Studies (INSEE). Peri-urbanization refers to the space between the city and countryside and is defined as a structure that occurs in the process of urban marginalization. It is well illustrated as a landscape interface between the city and countryside through interactions with rural or mountain villages.

Cluster IV was an area with excellent forest resources, such as FA and F5, among ecological values, with the highest RF, SC, and TR values. However, the LI and MI values were the lowest, and the BA and POP values were also low. This indicates that the regions included in Cluster IV are located in the forests adjacent to the protected area in the two mountain forest networks (NN and ND) or where forests are dominant and resource-rich, such as the upper region of ND and the region to the left of NN. Based on these forest resources, it was confirmed that there are many scenic and tourist spots, such as temples, valleys, and waterfalls.

Villages classified in Cluster V showed the lowest PA and NF values, confirming that the functional aspects of forest protection and conservation were low. In addition, the LI and MI values were lower than those of other clusters. It was confirmed that the villages included in Cluster V were located relatively far from the city center and adjacent to the forest. In contrast, as the TF, WF, and RF values of the economic value were high, the functional aspect of the forest showed strong characteristics. It can be classified as a village where forestry-based economic activities are active. In particular, it was concentrated in the central region of the NN.

Villages in Cluster VI showed the highest PA values among the ecological values, and the FA, F5, and AG values were also high. Several legally protected areas, such as natural parks, natural environment conservation areas, and wildlife protected areas, have been designated within the area, and well-preserved forest areas have increased as they

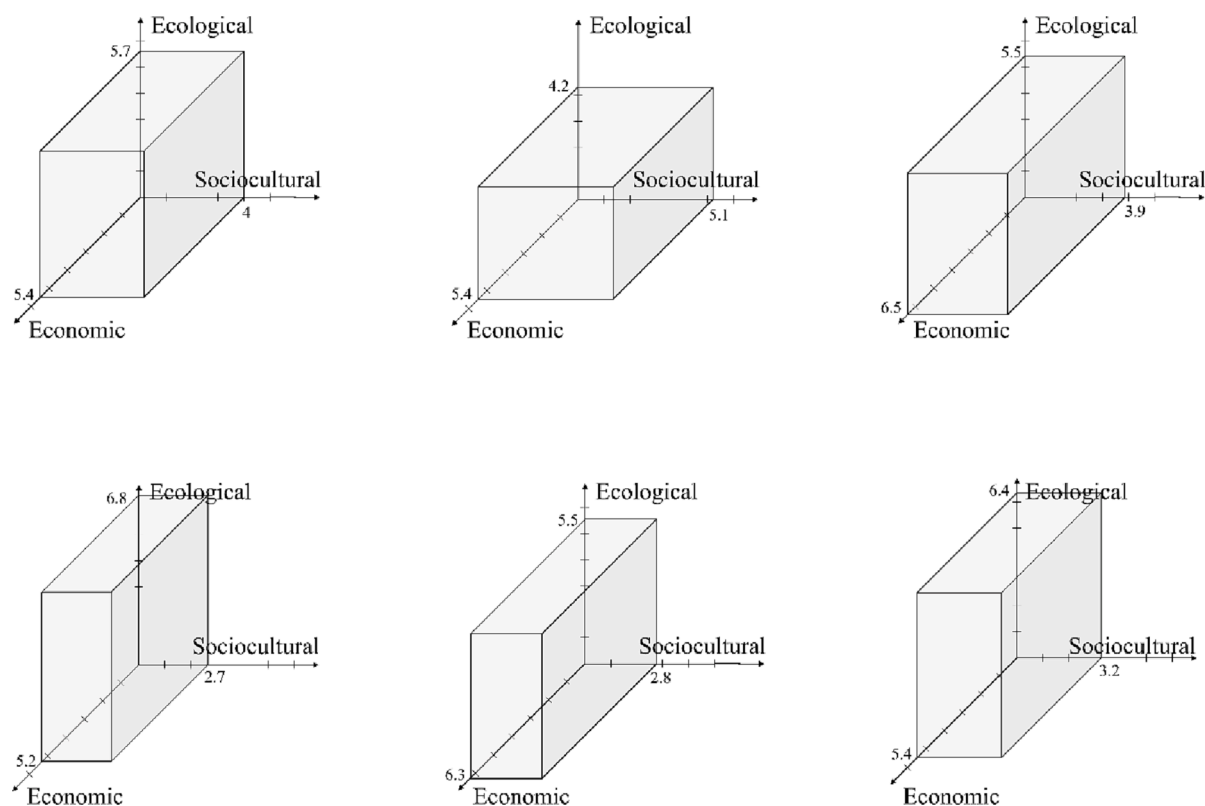


Fig. 6. Polygonal graphs for visualizing the distribution of ecological, economic and socio-cultural indicators in clusters a) I, b) II, c) III, d) IV, e) V, and f) VI using self-organizing map.

have expanded up the ND. In contrast, among the six clusters, the POP, BA, and CU values were the lowest, and the villages classified as Cluster VI were located farthest from the city. This is an area in Korea where current problems due to population decline and slow urbanization are emerging. However, as human intervention is minimized, forests are well conserved as areas with a high potential for protection.

3.3. Classification using Geo-SOM

The Geo-SOM algorithm, without a categorical group and with a map size of 17×12 , was trained using the Geo-SOM suite (Henriques et al., 2009). A U-matrix heatmap with the same number of clusters (six), as applied in the SOM and component plans (property plot) of each variable, is illustrated in Fig. 7. The grids were counted from top left to bottom, and the final grid ended in the bottom-right cell. The default color of the map is red for maximum values and dark blue for minimum values.

The U-matrix heatmap displayed in the Geo-SOM output visualizes the relative distance between neurons, and the difference in color are used to display the calculated difference between adjacent neurons. The blue area of the Geo-SOM plots represents the close distance between the vectors in the space of the input data, while the red area represents a large distance. It is possible to infer where the variables are distributed through heatmaps, depending on each variable. As a result of reflecting geographic location, values such as forest area, above five age class forest area, and timber production forest area are skewed toward the upper area of the ND. Moreover, BA, LI, and MI values were skewed to the lower area of ND and to the right of NN (Fig. 7).

Through the application of the Geo-SOM to map geo-reference data, spatially distinct villages around the forest were derived. Among the six clusters, Cluster IV had the highest number of villages (115), followed by Clusters III (101), II (98), I (38), V (16), and VI (11). The number of villages included in the cluster showed biased results when the spatial attributes were assigned. As a result of classifying variables into three categories, the region of Cluster II showed the strength of the ecological, Cluster I showed economic, Cluster IV showed socio-cultural, and clusters III, V, and VI exhibited complex characteristics (Fig. 8).

Cluster I was geographically distributed around the ND midpoint and included most of the Gyeongju City and Yeongcheon County. In Cluster I, the PA, AG, and TF values were the highest, and the NF, SC, and TR values were the lowest. The statutory protected area was small, but most of the other areas appeared to be designated as the first area of the ecological nature map, and agricultural land was distributed around the forest. The first area of the ecological nature map is an area with high ecological value, such as a habitat of endangered species, excellent landscape, and a habitat of representative major vegetation

communities, which cannot be developed, suggesting that the area in Cluster I is ecologically superior.

Geographically, Cluster II was broadly classified up to the start of the ND mountain range above the Cluster I region. The ecological values of FA and F5 were the highest, and those of WA, POP, and BA were the lowest. Cluster II was an area with particularly developed forests in the ND mountain range, located far from the city, with problems caused by a decrease in the workforce population size. The SOM analysis showed characteristics similar to those of Cluster VI.

Cluster III was located from the left to the middle region of the NN, at the end of the Baekdudaegan Mountain Range. WA and WF values were the highest, while PA and LF values were the lowest. It is presumed that the PA value is low because there are many sections where forests are damaged or cut off within most of the clustering range, even though Jiri Mountain National Park, which is a statutory protected area located in this region. Data verification indicated that most of the valleys and reservoirs are distributed in Cluster III, which is classified as an area with high water source potential.

Cluster IV is located at the bottom of the ND and to the right of the NN. Among the sociocultural values, the POP, BA, LI, and MI values were high. In contrast, the FA, AG, RF, and WT were the lowest, and other forest-related variables were generally low. Based on the SOM results, Cluster IV shared characteristics with Cluster II, and included metropolitan areas such as Busan, Gimhae, and Changwon, which have high population densities. This area has experienced the most severe forest damage and disconnection owing to industrial complexes and residential areas.

Cluster V was classified as the region adjacent to Cluster VI. The WA, NF, and PA values were high, even in areas adjacent to large cities and a number of protected areas, such as protected water source areas, protected wetland areas, natural parks, and genetic forest resource reserves, were designated. Moreover, Cluster V showed that the forests in the region are well preserved enough to be called the Alps of Korea. However, because it is adjacent to a large city, the BA and POP values were high. Additionally, the AG and LF values were high, confirming the complex characteristics of a region located in the outskirts of a metropolitan city.

Cluster VI was located between clusters V and I. Overall, the FA value was high along with the RF, LF, CH, and SC values, indicating that the use and access of forests were relatively free. In contrast, F5 and WF values were the lowest, indicating that the trees in the forest were not too old; LI and MI were also the lowest, indicating that the villages in Cluster VI were comparably far from cities. Cluster VI confirmed that there were several protected areas, such as natural parks, ecosystems, landscape conservation areas, and genetic forest resource reserves. Based on these superior forest resources, several campsites, pensions,

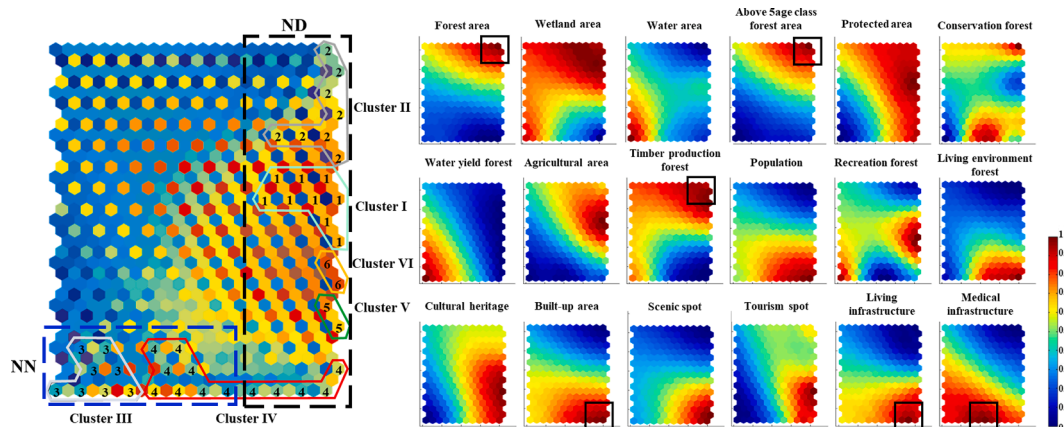


Fig. 7. Geographic self-organizing maps (Geo-SOMs). (a) Heatmap with clearly identified clusters in NN: Naknam-Jeongmaek and ND: Nakdong-Jeongmaek regions. (b) Component (property map) plans.

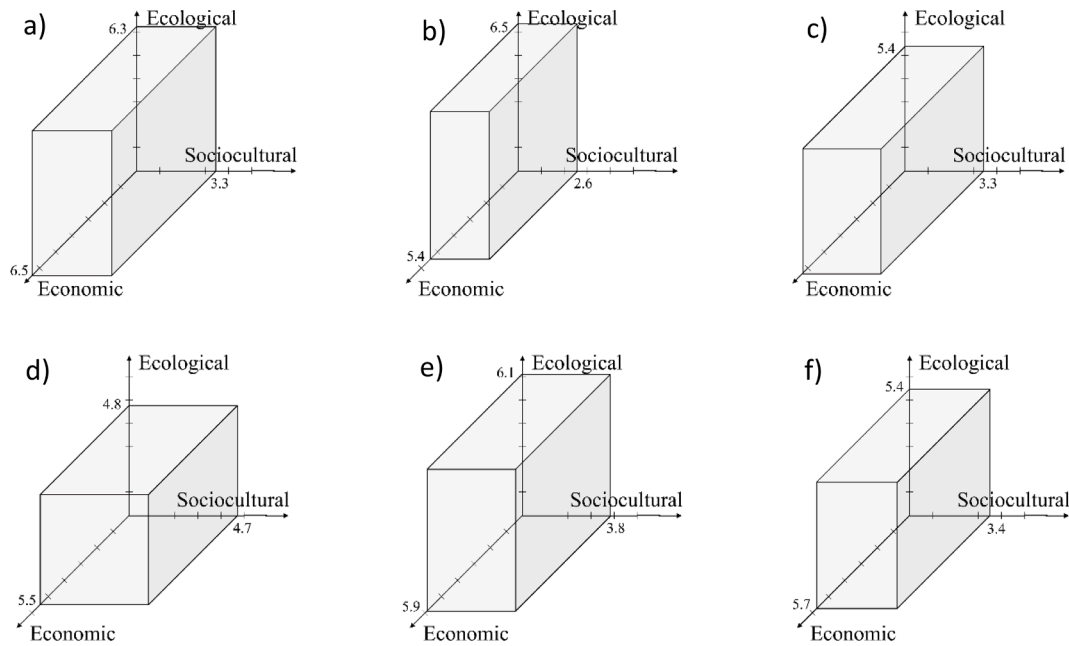


Fig. 8. Polygonal graphs for visualizing the distribution of ecological, economic, and socio-cultural indicators in clusters a) I, b) II, c) III, d) IV, e) V, and f) VI using a geographic self-organizing map.

and accommodations are distributed in this area.

4. Discussion

Several socio-economic and socio-cultural studies have been conducted on various aspects of ecological systems such as agroforestry (Murthy et al., 2016; Pandey et al., 2017; Nath et al., 2021; Kumar et al., 2021a, Kumar et al., 2021b), ecosystem management (Das et al., 2022b, Das et al., 2022a), climate change, (Pandey et al., 2017, Gupta et al., 2019; He et al., 2021; Kumar et al., 2021a, Kumar et al., 2021b) and

sustainable development goals (Hák et al., 2016). In this study, it was confirmed that the villages showed distinctive spatial distribution patterns reflecting the unique characteristics of each variable examined using SOM and Geo-SOM techniques in terms of the socio-ecological, economic, and cultural systems for forest management. In particular, the characteristics of forested and urbanized areas were clearly distinguished, which can be considered an important basis for preparing a potential forest management plan and for conservation. Cluster analysis applying only the characteristics of a variable can show spatially discontinuous results; however, it can also produce clear results on

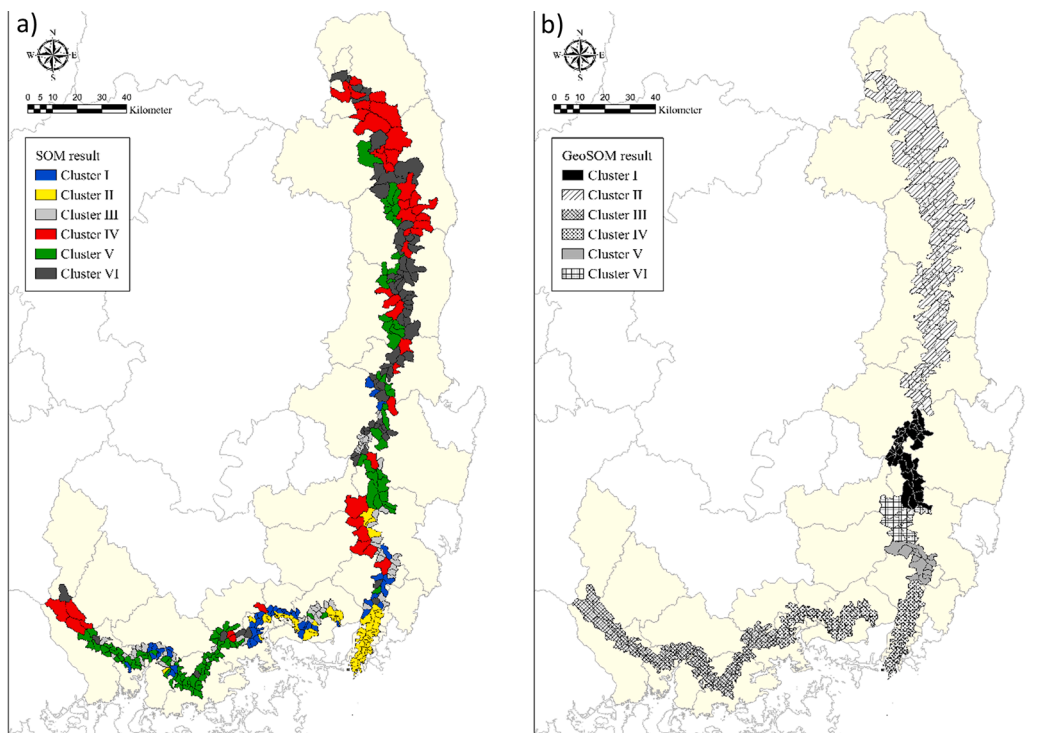


Fig. 9. Spatial distribution of villages in (a) self-organizing map (SOM) and (b) geographic self-organizing map (Geo-SOM) clusters.

which variables are affected. This study applied spatial characteristics to distinguish between areas with special cluster characteristics. This could be a problem because the clustering result applying only the characteristics of the variable cannot be clearly interpreted for setting the scope of the management target. However, this study proved that it could be proposed as more objective and reliable evidence because the analysis results through SOM and Geo-SOM synthesize the attributes of spatial characteristics with attributes of clustering (Fig. 9). In addition, in future management and policy proposals, a divided cluster can be named, which reflects the characteristics of each cluster (Shon and Youn, 1997; Kim et al., 2020; Ko et al., 2020).

Selecting the map size is crucial when using SOM (Moosavi, 2017). The optimal grid size can be determined by checking the QE and TE for each node and increasing the node value incrementally, even if the same node is configured. This study examined the optimal size of the SOM with the integrated effect of error deviation for the same grid size. Further validation of the size estimation approach used in this study is required.

Jeong and Kim (2019) noted that the low frequency of the appearance of a cluster characteristic in the clustering process was due to the structural problems of the data. Spatial visualization of the cluster by SOM did not show geographical distribution (Fig. 9a). Geo-SOM is an effective tool for analyzing spatial data patterns (Fig. 9b); however, biased results may occur when the data and target sites are large. This highlights the importance of dividing the data into regional and detailed

units when setting the scope and plan for forest management.

In this study, we observed a close but biased relationship between SOM and Geo-SOM, according to spatial characteristics (Table 2). Cluster I of the Geo-SOM is a village included in the midpoint of ND and shows strong ecological characteristics. In the SOM results, it was found that clusters I and II with high sociocultural values did not apply, which are large cities or areas adjacent to large cities. This area requires a management plan to improve accessibility to large cities and introduce infrastructure while conserving forests by minimizing development. Geo-SOM Cluster II villages are located above ND and contain more than half of the ND mountain range. The region that is mostly affected by forest resources has the lowest population density and urbanization. Clusters II and III belong to urban areas and were not included in the SOM results. Considering that the current problem of workforce population decline is serious, the appropriate use of forest resources that can increase access to large cities and generate economic income is necessary. Geo-SOM Cluster III was a village around the midpoint of the NN. It is an area with a strong water resource function of the forest and mainly overlaps with Cluster V of SOM. Geo-SOM Cluster III shows strong ecological characteristics, but in the area where the city is formed, the forest is damaged and cut off. Therefore, it is necessary to prepare a management plan to secure the connectivity of the forest. Geo-SOM Cluster IV is a typical metropolis; although it is distributed in all clusters of SOM as a region, it shows strong characteristics in sociocultural values such as BA, CH, LI, and MI. In terms of location, it was formed around Busan, the largest metropolitan area in the two forest networks. The area is characterized by high population density and development pressure, making simultaneous efforts to protect and restore degraded and disconnected forests a priority. Although Geo-SOM and SOM Cluster V are located near large cities, they have abundant forest resources and a strong *peri-urban* character as many protected areas are designated. Finally, Geo-SOM Cluster VI appears partially in ND. It has characteristics similar to Cluster V of SOM, indicating that the tourism industry centered on forest resources has developed by utilizing accessibility to the city.

Therefore, this study aimed to categorize villages adjacent to the two forest networks that have formed the original culture and society for an extended time to manage forests according to the characteristics of their resources. This differs from previous studies on the classification of rural, mountain village, and landscape types of village units (Lee, S. and Lee, 2011; Park and Kang, 2016; You et al., 2017). Continuous data updates and research from various perspectives is necessary to foster socio-ecological, economic, and cultural factors of villages within the forest network by connecting interior and exterior environments.

Monitoring the natural environment and identifying dominant indicators facilitate the formulation of sustainable management policies (Salas-Garita and Soliño, 2021). This study used geographic information system-based spatial data to examine the characteristics of villages using dominant indicators. In addition, despite the lack of landscape and resource research on forests, it is meaningful that the characteristics of each forest are derived and types are classified. In addition, the village type that reflects the spatial characteristics of the Geo-SOM result has a broad classification characteristic in setting the scope of forest management, and the SOM result is expected to enable detailed management plans linked to cluster characteristics.

However, this study has significant limitations. Regarding variable selection, the contribution of the variables and their collinearity checks were not conducted. We used a 1 km buffer from the ridge line; if the buffer area changed, the result would change. Therefore, the optimal buffer zone should be examined and analyzed using the SOM approach. We analyzed the QE and TE to examine the optimal map size; other errors, such as the Kaski-Lagus error and percentage explained variance metrics, should also be applied. The optimal map size and cluster number for SOM, which could be examined independently, were used for Geo-SOM analysis. We analyzed only 18 features; many indicators, such as climatic and anthropological, were not considered. Currently,

Table 2
Average of variables by cluster.

Indicators	Variables	Cluster SOM/Geo-SOM					
		I	II	III	IV	V	VI
Ecological value	FA	4.44/	3.15/	2.91/	8.76/	5.83/	8.33/
		5.71	8.45	4.87	3.43	4.63	7.18
	WT	5.24/	2.52/	7.7/	4.8/	6.73/	5.33/
		6.11	6.09	6.21	4.04	6.94	5
	WA	5.78/	3.29/	7.82/	4.63/	6.72/	4.17/
		7.42	4.18	7.53	4.22	5.75	5.09
	F5	4.98/	3.65/	3.98/	8.78/	5/	7.76/
		7.05	7.69	4.22	4.51	5.56	2.64
	PA	5.34/	6.95/	5.79/	6/6.1	4.35/	7.11/
		7.34	6.15	4.42		6.69	5.64
	NF	8.44/	5.68/	4.96/	7.88/	4.28/	5.67/
		4.13	6.26	4.88	6.48	7.31	6.73
	AG	7.38/	4.08/	5.93/	5.24/	6.21/	4.77/
		4.37	5.03	7.38	5.36	4.06	3.45
Economic value	WF	3.94/	3.75/	9.02/	4.66/	5.49/	5.91/
		8.21	5.7	5.29	4.27	7.38	6.45
	TF	3.06/	4.38/	4.16/	4.34/	7.99/	7.14/
		7.11	6.47	6.59	3.99	3.13	4
	RF	7.26/	9.11/	7.02/	4.83/	5.14/	3.55/
		5.34	3.9	5.57	8.38	7.75	5.91
	POP	5.46/	5.43/	6.32/	7.15/	6.62/	5.64/
		7.24	5.69	6.58	5.27	7	8.82
	LF	5.54/	7.74/	7.28/	6.61/	5.99/	6.06/
		7.53	6.23	5.6	6.89	7.69	7.91
	BA	6.8/	9.06/	7.37/	2.88/	4.6/	2.39/
		4.55	2.32	5.36	8.43	7.25	5.36
	CH	0.43/	1.58/	0.53/	1.15/	0.41/	0.25/
		0.2	0.4	0.54	1.09	1.12	1.32
Socio-cultural value	TR	0.5/	0.84/	0.52/	1.48/	0.63/	0.69/
		0.71	0.8	0.63	0.76	0.69	1.05
	SC	0.19/	0.79/	0.55/	1.17/	0.31/	0.21/
		0.27	0.4	0.52	0.57	0.44	0.86
	LI	7.4/	7.62/	5.63/	2.39/	4.28/	5.88/
		4.66	3.46	6.09	7.54	4.38	2.91
	MI	7.18/	8.17/	5.35/	3.07/	3.45/	6.59/
		5.42	4.65	4.26	7.7	4.88	4.09

FA: forest area; WT: wetland area; WA: water area; F5: above 5age class forest area; PA: protected area; NF: Natural environment conservation forest area; AG: agriculture area; WF: water yield forest area; TF: timber production forest area; POP: productive population; LF: living environment forest area; BA: built-up area; CH: cultural heritage; TR: tourism spot; SC, scenic spot; LI: distance from living infrastructure; MI: distance from medical infrastructure.

climate change vulnerability and adaptation are challenging (Pandey et al., 2017; Gupta et al., 2019; He et al., 2021). Understanding the significant influencing factors in each cluster could further ease policy making for micro management (Salas-Garita and Soliño, 2021; Ndao et al., 2022) but this study only clustered and highlighted the dominant indicators of each cluster. The application of econometric and ecological distribution models to examine the driving properties of each cluster and village network was also missing in this study. Overcoming these limitations and developing optimal management measures is warranted in future studies.

5. Conclusion

In this study, the characteristics of each village (administrative district) adjacent to the two mountain ranges on the Korean Peninsula were analyzed by mapping various resources and variables distributed within each target area using SOM and Geo-SOM. The current approach to characterizing and clustering villages near mountain forests is a new method for supporting the establishment of forest management plans. This study is useful given the limited data availability and application of remote sensing and geographical information systems. However, classifying numerous databases constituting the target site is considered an adequate method for classifying non-continuous data. The results are expected to provide a general evaluation of landscape management, such as forests and mountains, and more efficient urban or rural management plans.

CRedit authorship contribution statement

Tae-Su Kim: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Data curation, Writing – review & editing. **Thakur Dhakal:** Formal analysis, Writing – original draft, Methodology, Software, Writing – review & editing. **Seong-Hyeon Kim:** Data curation, Methodology, Software, Writing – original draft. **Ju-Hyoung Lee:** Supervision, Visualization, Validation, Writing – review & editing. **Su-Jin Kim:** Funding acquisition, Supervision, Investigation, Validation. **Gab-Sue Jang:** Conceptualization, Supervision, Investigation, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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